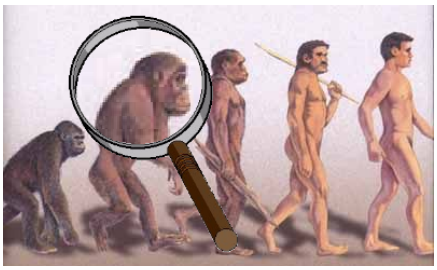
Computational Biology

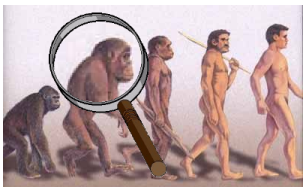
Lecture 11: *Population Genetics*



Fisher-Wright model, ODEs, Markov chains.

Outline

1. **Population Genetics**
2. Facts of Life
3. Infinite Population Model
4. Finite Population Model



Evolution and Molecular Biology

- Evolution is the theory that underpins our understanding of biology
- Molecular biology has open up a whole new perspective on evolution as we see similar proteins in related species
- This evolution ancestry helps us understand molecular biology as similar proteins usually share the same function
- However we can also use molecular biology to help understand evolution—more of this when we discuss phylogenetic tree
- But how does evolution work?

Evolutionary Population Dynamics

- Since as least the 1920's researchers have tried to model evolution
- The field is known as **population genetics**
- Much of it was developed before we understood the mechanisms of heredity
- It is therefore rather abstracted from biology
- However it was enormously influential to our understanding of evolution

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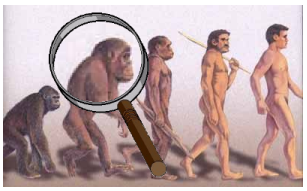
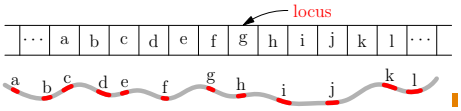


Fisher-Wright Model

- The classic model of population genetics is the Fisher-Wright (or Wright-Fisher) model developed independently by Fisher and Wright around 1930
- It is important because it laid the theoretical framework for many of the prejudices held today
- In its simplest form it is a model of what happens at a single locus (site) under selection, mutation and crossover



Locus and Alleles

- The locus can be thought of as the location of a gene
- The type of gene at a site will differ between individuals (e.g. one for smooth peas and one for wrinkled peas)
- The different types of genes are known as **alleles**
- We assume that genes can be modified from one allele to another by mutation
- Alleles are also exchanged during sexual reproduction (crossover)

Fitness

- Fitness has a technical meaning in population genetics; it is a measure of the expected number of offspring an individual produces
- If one individual is 10% fitter than another it will (in expectation) produce 10% more offspring
- Of course, the individual may get flattened by a car before it reaches sexual maturity and never produce any offspring
- Thus fitness measures potential to reproduce

Epistatic Interaction

- Many genes interact (we now know they often code for proteins that directly interact chemically in many cases)
- Thus the fitness of an allele will typically depend on other alleles at other locations
- This interaction is known as **epistasis**
- To keep things simple in the Fisher-Wright model we assume no epistasis
- There exists more complex models which treat epistasis between two or more loci, but these rapidly become rather complex

Sex

- Sex is surprisingly ubiquitous in higher organisms
- Few higher organisms seem to reproduce asexually (although even if they do, many sometimes also reproduce sexually)
- Asexual higher organisms all would have produced sexually sometime in their past
- Why sex is surprising is that if females produce asexually (producing clones) then they breed at twice the rate of mixed couples—hence they are twice as fit
- Sexual reproduction has to give at least a two fold advantage in fitness on a very short time scale to survive

Linkage Equilibrium

- As well as no epistasis the model assumes lots of crossover (compared to selection and mutation)
- This is not unrealistic in many cases
- Under this assumption we can treat the loci independently and concentrate on a single locus
- This is known as **linkage equilibrium**
- We therefore only need to consider the proportion of the different alleles at the locus
- In an asexual population, the alleles at different sites are coupled and the situation is much harder (in fact, unsolved)

Diploidy

- Most organisms are diploids, that is they carry two copies of DNA; one from their father and one from their mother
- Like sex, it is not altogether clear what advantage diploidy brings
- Clearly, diploids can tolerate mutations that make one set of genes dysfunctional
- But, this also allows detrimental mutations to build up in a genome
- Incidentally, mating between close relatives is likely to create nonviable offspring because it is likely to combine two dysfunctional alleles

Haploidy

- To keep things simple we are going to assume each individual only carries one set of genes
- Organisms that do this are said to be **haploidy**
- There are various organisms (e.g. male bees, wasps, ants, many fungi) that are haploid
- Generalising to diploid organisms is just fiddly—it mainly involves adding in factors of two every so often, but it does not substantially change things
- Incidentally many plants are polypoids having more than two copies of each gene

Gender

- Life tends also to be gendered
- This possibly just reflects that finding a mate consumes energy and by specialisation some energy can be saved
- However, this is another complicating factor, which adds little to the model
- I will assume a **monoecious** population, i.e. one with only one gender, although it still mates
- These are all simplifying assumptions, but it allows us to concentrate on the role of selection, mutation and crossover

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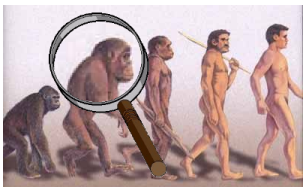


Single Locus Model

- We consider the case where the allele can take just two states A and B
- We call B the mutant
- We make the following assumptions
  - B has a selective advantage of $1 + s$
  - A mutates to B with a forward mutation rate of u
  - B mutates to A with a backward mutation rate of v
  - The population size, P, is unchanged from generation to generation

Many Seeds

- We consider the population changing from one generation to the next by a three stage process
  - Each individual produces a very large number of seeds in proportional to its fitness (selective advantage)
  - The seeds are mutated according to the mutation probabilities
  - We select a sample of P of the seeds independently to form the next generation
- Initially we assume P is infinite and consider only the proportion of mutants in the population

Dynamics

- Assume we start with n mutants
- After selection we have a proportion p_s of mutant seeds

$$p_s = \frac{(1+s)n}{(1+s)n + P - n} = \frac{(1+s)n}{P + s\,n}$$

- After mutation we have a proportion p_{sm} of mutants

$$p_{sm} = (1-v)p_s + u(1-p_s) = \frac{(1-v)(1+s)n + u(P-n)}{P + s\,n}$$

Infinite Population Approximation

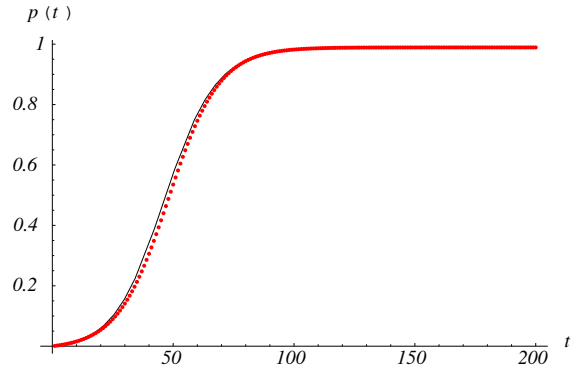
- If we assume an infinite population then this will be the proportion of individuals at the next generation

$$p(t+1) = \frac{(1-v)(1+s)p(t) + u\big(1-p(t)\big)}{1 + s\,p(t)}$$

- If the effect of selection and mutation is small then we can approximate the dynamics by a differential equation

$$\frac{\mathrm{d}p}{\mathrm{d}t} \approx p(t+1) - p(t) = \frac{\big(u - (u+v-s+u\,s)p(t) + s\,p^2(t)\big)}{1 + s\,p(t)}$$

Infinite Population Prediction



$s = 0.1,\ u = v = 0.001$

Difficulties of Infinite Populations

- In an infinite population a mutation (even if it is very unlikely) will always occur in a small fraction of individuals
- Selection can rapidly amplify the mutants if they are fitter—note fitter individuals initial grow exponentially

$$(1+s)^t = \mathrm{e}^{t\,\log(1+s)} \approx \mathrm{e}^{s\,t}$$

- If the mutation rate is small then in finite populations the mutant may take a very long time to appear
- Thus infinite populations can have far more rapid take over times than is experience in large, but finite, populations

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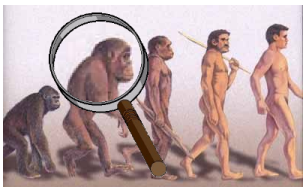


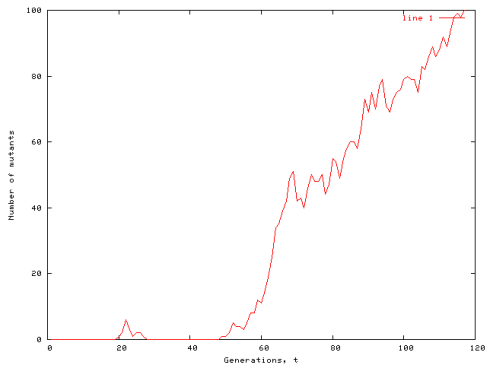
Finite Populations

- To model finite populations, we have to draw a sample of P individuals from the seeds
- The probability of drawing $n' = n(t+1)$ mutants given a proportion $p_{sm}(n(t))$ of mutants in the seed population is given by a binomial distribution

$$\mathbb{P}(n') = \binom{P}{n'} p_{sm}^{n'} \left(1 - p_{sm}\right)^{P-n'}$$

- (Recall that p_{sm} is the proportion of mutant seeds after selection and mutation and depends on $n(t)$)
- We can simulate the evolution of a population, by drawing a random binomial deviate at each generation

Finite Population Run



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Markov Models

- The evolving population can be modelled as a Markov chain
- The population at time t depends only on the population at time $t-1$
- It is *conditionally independent* of the population at earlier times *given* the population at time $t-1$
- We define $p_n(t) = \mathbb{P}(n(t) = n)$ to be the probability that there are n mutants at time t
- Let $W(n \to n')$ be the transition probability of a population with n mutants going to a population of n' mutants in the next generation

$$p_{n'}(t+1) = \sum_{n=0}^{P} W(n \to n') p_n(t)$$

Matrix Formulation

- Denote the *probability vector* $(p_0(t), p_1(t), \ldots, p_P(t))$ by $\boldsymbol{p}(t)$
- Let $\mathbf{W}$ be the *transition matrix* with elements $W_{n'n} = W(n \to n')$
- The dynamics can be written in matrix form

$$\boldsymbol{p}(t) = \mathbf{W}\boldsymbol{p}(t-1) = \mathbf{W}\left(\mathbf{W}\boldsymbol{p}(t-2)\right) = \mathbf{W}\mathbf{W}\cdots\mathbf{W}\boldsymbol{p}(0) = \mathbf{W}^t\boldsymbol{p}(0)$$

- Where $\boldsymbol{p}(t=0)$ encodes our *initial conditions*
- If initially $n(t=0) = 0$ then $p_0(t=0) = 1$ and $p_i(t=0) = 0, \forall i > 0$

Markov Analysis

- In our problem the transition matrix, $\mathbf{W}$, has elements

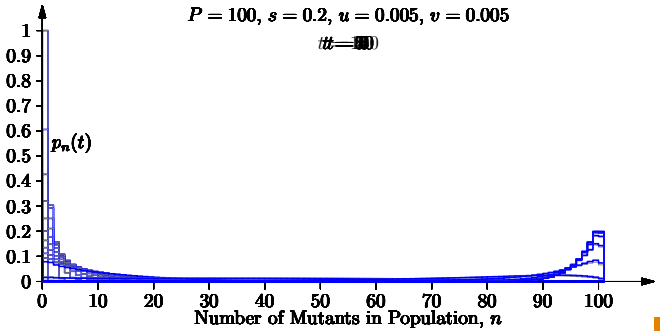
$$W_{n'n} = \binom{P}{n'} \left(p_{sm}(n)\right)^{n'} \left(1 - p_{sm}(n)\right)^{P-n'}$$

- where $p_{sm}(n)$ is given by

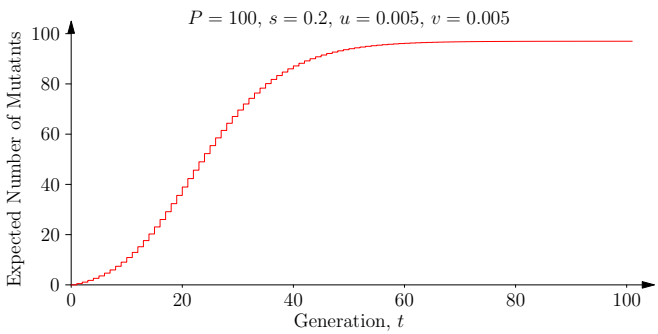
$$p_{sm}(n) = \frac{(1-v)(1+s)n + u(P-n)}{P + s\,n}$$

- This provides an exact formulation of a finite population

Example of Markov Analysis



Mean Number of Mutants



Problems with Markov Chain Analysis

- The Markov Chain analysis is exact and provides a full picture of the probability of the population being in a certain state
- Unfortunately, the transition matrix $W(n \to n')$ is of size the number of states squared
- The number of states for a population of size P is $P+1$
- For a large population writing the transition matrix becomes infeasible
- Even when it is feasible we don't get much intuition on changing the parameters

Conclusion

- The Fisher-Wright model provides a simple model undergoing selection, mutation and crossover
- It allows us to explore the effects of selection pressure and mutation rate
- We can solve the infinite population model directly or by solving an ordinary differential equation (ODE)
- Unfortunately infinite population models don't always capture the behaviour of finite populations even when the population is very large
- We can simulate finite population models
- For small populations we can use Markov models